Block GIT-207

Gastro-Intestinal System

Drug Treatment of Peptic Ulcer Disease

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Faculty of Medicine

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Drug Treatment of Peptic Ulcer Disease

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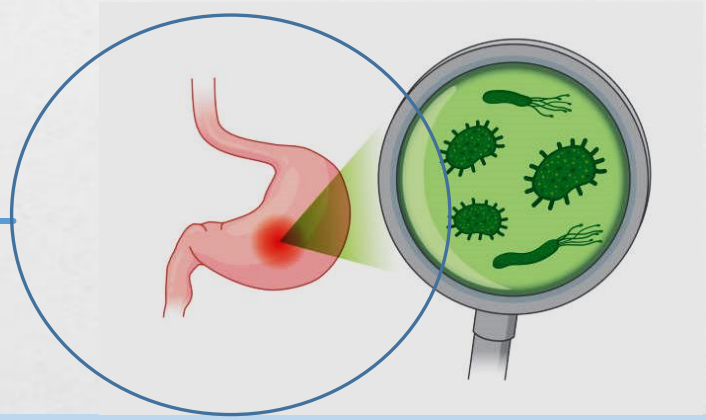
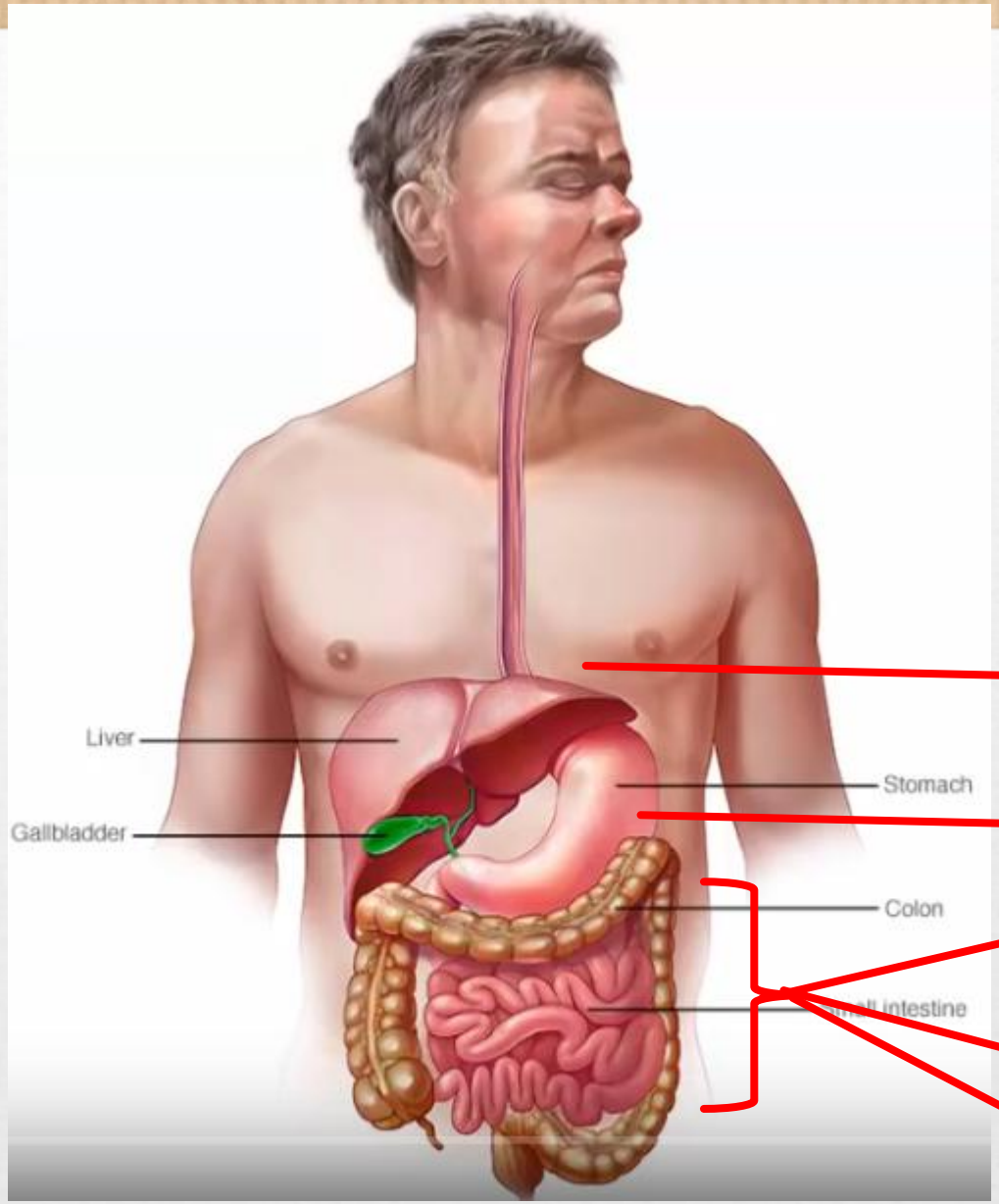


ILOs

- **Treatment of Peptic ulcer**
  - Anti-secretory Drugs
  - Mucosal protective Drugs
  - Antacids
  - Eradication of Helicobater pylori

REFERENCE BOOKS

- Lippincott and Illustrated Reviews, Pharmacology, 6th edition, 401-409
- Elsevier's Integrated Review Pharmacology, p17-178
- First Aid for the Basic Sciences Organ Systems, 3rd edition, p259-262



GERD

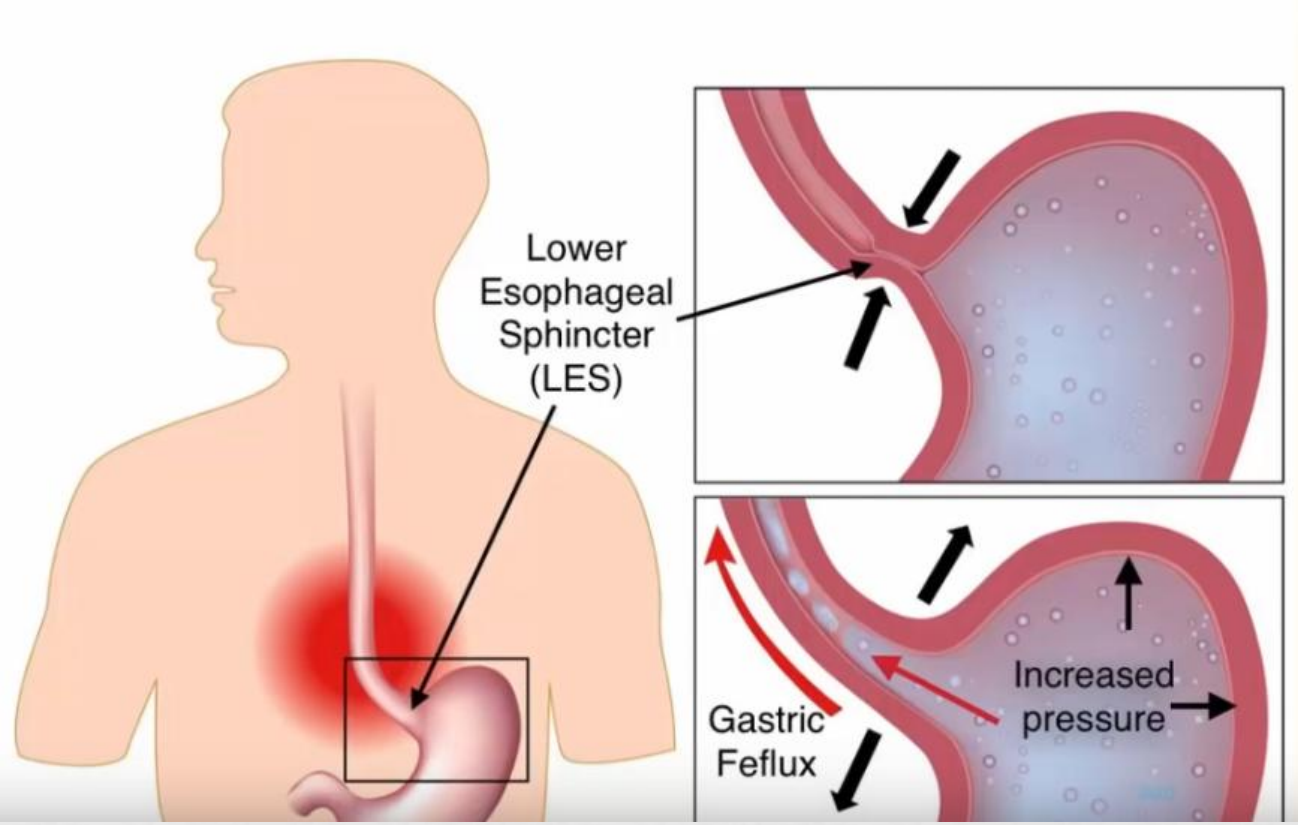
Peptic Ulcer

Motility Disorders
(Diarrhea, constipation)

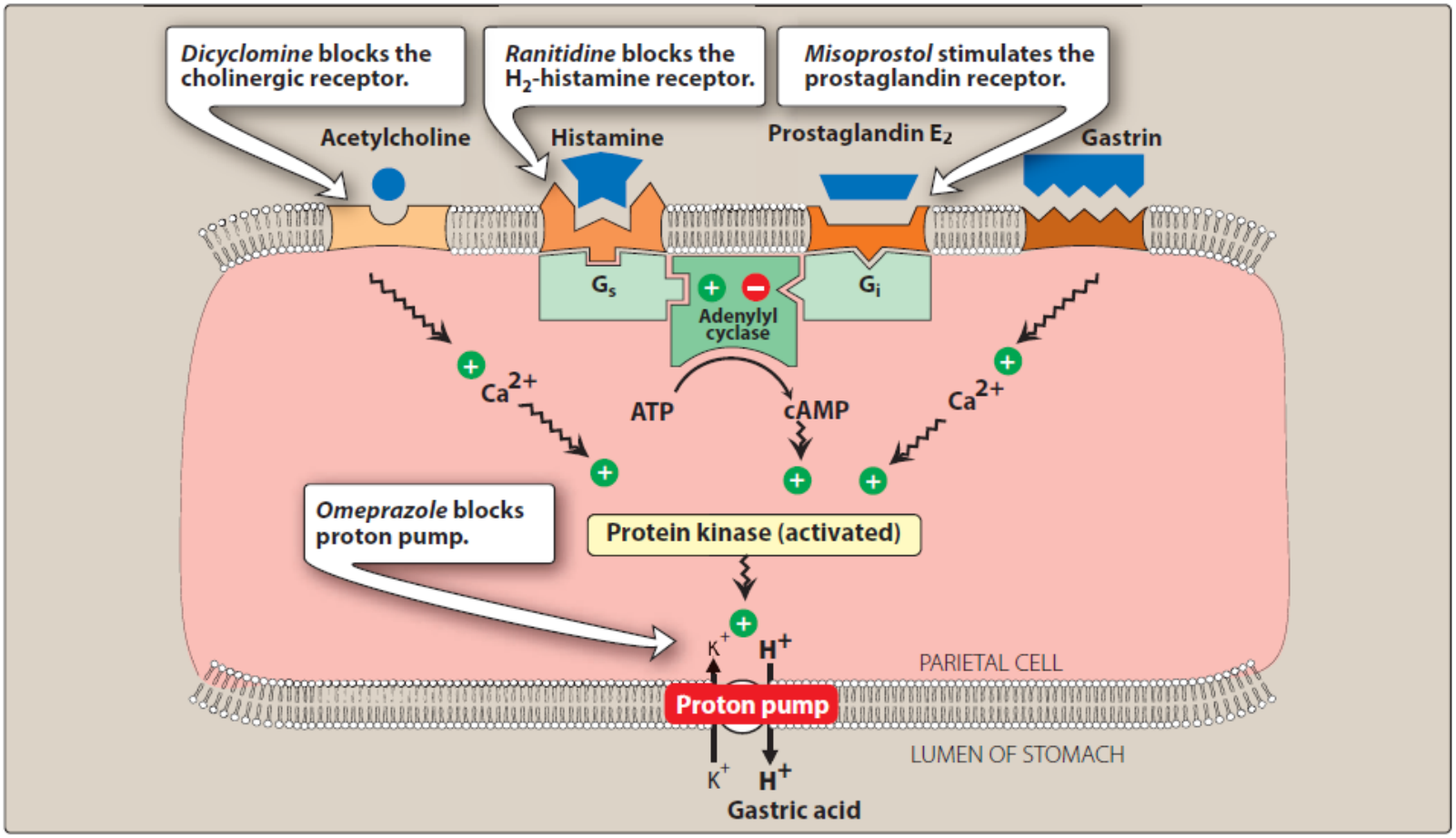
IBD

IBS

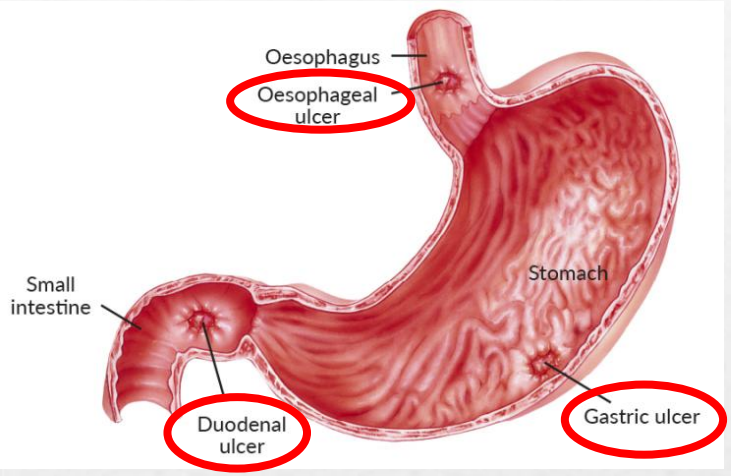
Gastroesophageal Reflux Disease (GERD)



Physiology of Acid Secretion

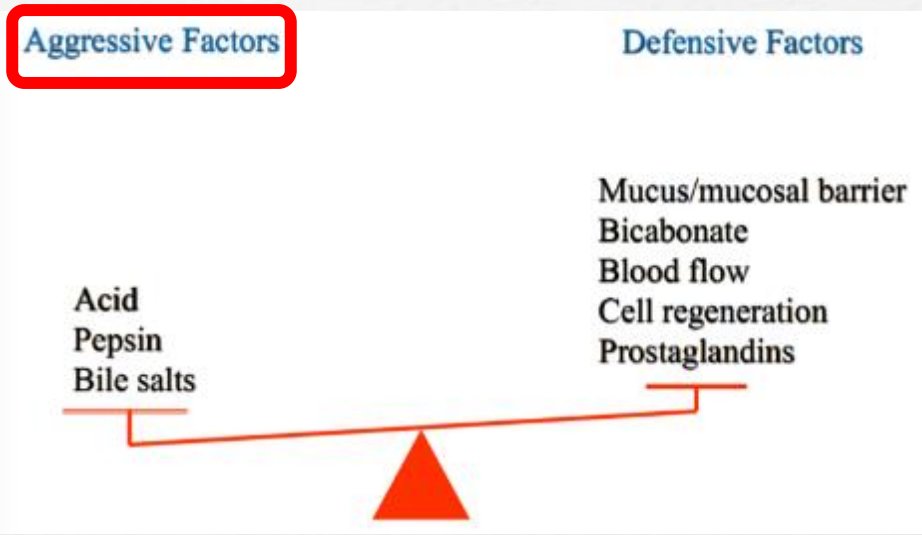


Peptic Ulcer



- A group of *ulcerative disorders of the upper gastrointestinal* (GI) tract that require *acid and pepsin* for their formation.
- They may occur at *stomach, duodenum* or lower part of the esophagus.
- It affects approximately 10% of the population of western countries.
- **The incidence of duodenal ulcer is four to five times higher than that of gastric ulcer (GU).**

Imbalance between aggressive factors and defensive factors.

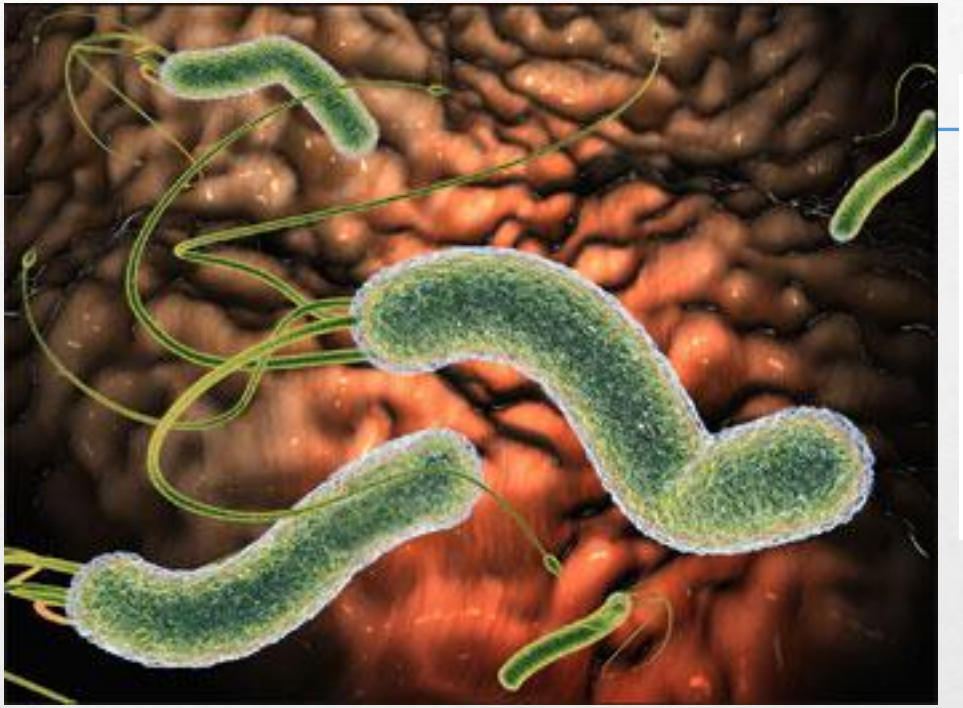


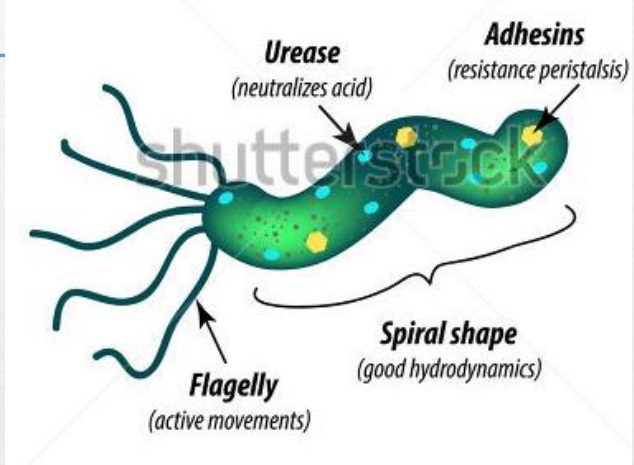
90% of peptic ulcers are caused by infection with the bacterium *Helicobacter pylori* or by *use of nonsteroidal anti-inflammatory* drugs (NSAIDs).

Pathophysiology

Mucosal erosions or ulceration arise when the effects of aggressive factors (**Acid**, **Pepsin**, **H. pylori**) overwhelm the defensive factors of the gastrointestinal mucosa (**Mucus** and **Bicarbonate** secretion, **PGs**, **Blood Flow**).

H pylori



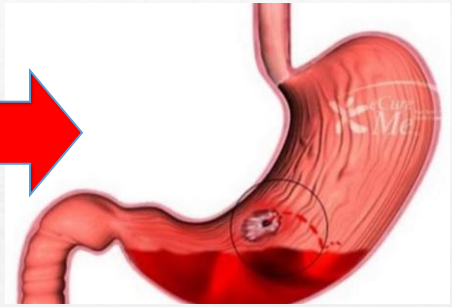

Urease
(neutralizes acid)
Adhesins
(resistance peristalsis)
Spiral shape
(good hydrodynamics)
Flagelly
(active movements)
shutterstock

Manifestations

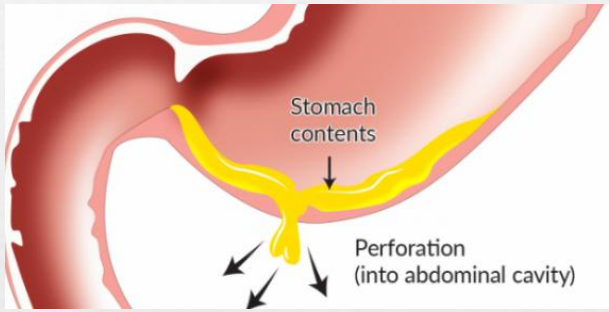
- **Epigastric pain** is the most frequent symptom that may be **burning, vague discomfort, fullness, or cramping**.
- It can be **nocturnal pain** that may awaken patients from sleep, especially between 12 am and 3 am.
- Pain from **duodenal ulcers** often occurs **1 to 3 hours after meals** and is usually **relieved by food (overweight)** , whereas **food may precipitate or accentuate ulcer pain in gastric ulcers (underweight)**.
- Heartburn, belching, and bloating often accompany pain.
- Anorexia, nausea and vomiting accompany gastric ulcer more than duodenal ulcer

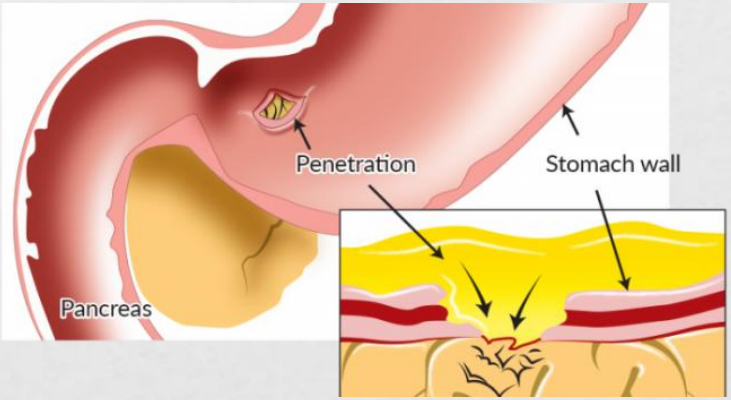
Complications

- **Upper GI bleeding** Bleeding may be occult or present as melena or hematemesis.

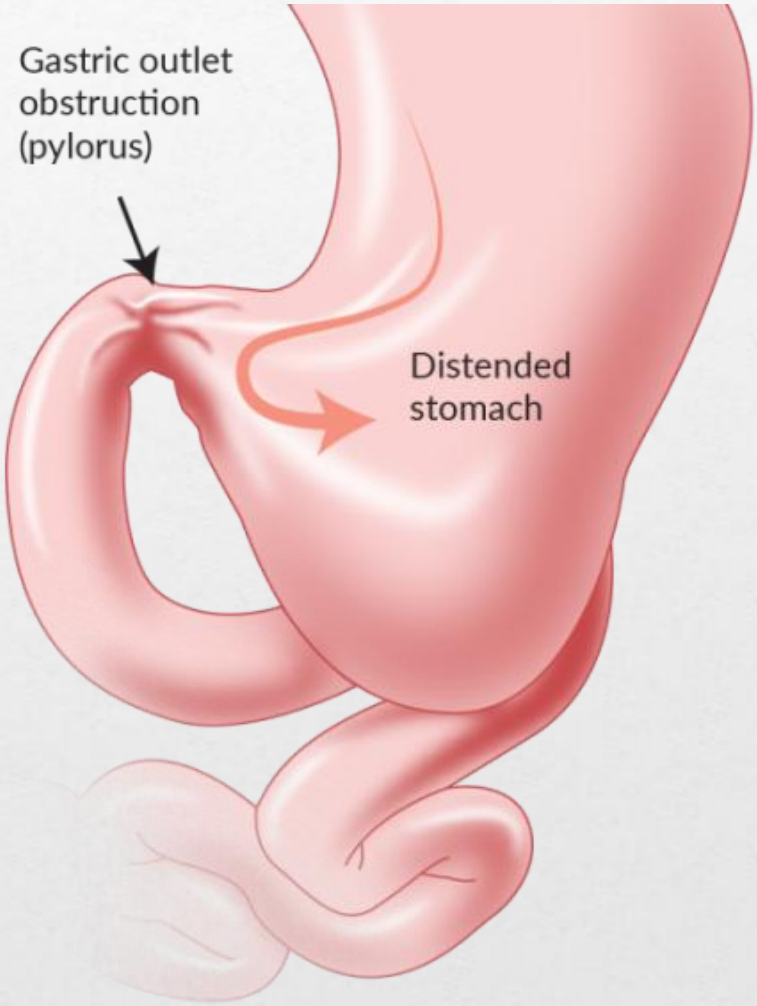


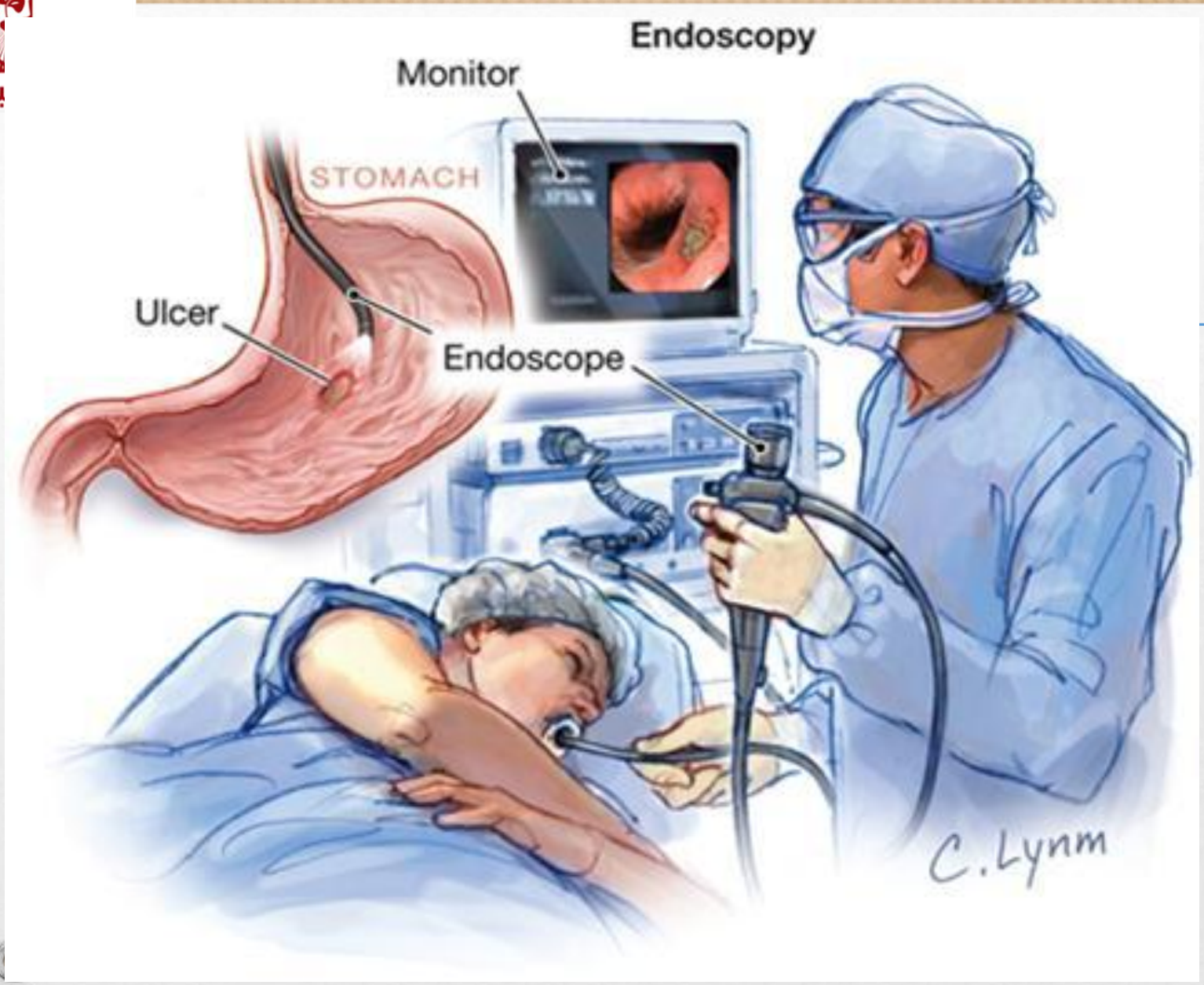
- **Perforation** is associated with sudden, sharp, severe pain, beginning first in the epigastrium but quickly spreading over the entire abdomen.
- **Penetration** into an adjacent structure (eg, pancreas, biliary tract, or liver).





- **Gastric outlet Obstruction** typically occur over several months and include early satiety, bloating,
- anorexia, nausea, vomiting, and weight loss.




Endoscopy
Monitor
STOMACH
Ulcer
Endoscope
C.Lynm

- **Gastro-esophageal reflux (GERD)** the acidic contents of the stomach enters into the oesophagus causing a burning sensation , indigestion and dyspepsia.

- **Zollinger-Ellison syndrome** is caused by tumor of gastrin secreting cells characterized by excessive secretion of gastrin that stimulates gastric acid secretion.

Principles of management

- Symptomatic relief
- Promotion of ulcer healing
- Prevention of recurrence
- Prevention of complications

I- General management

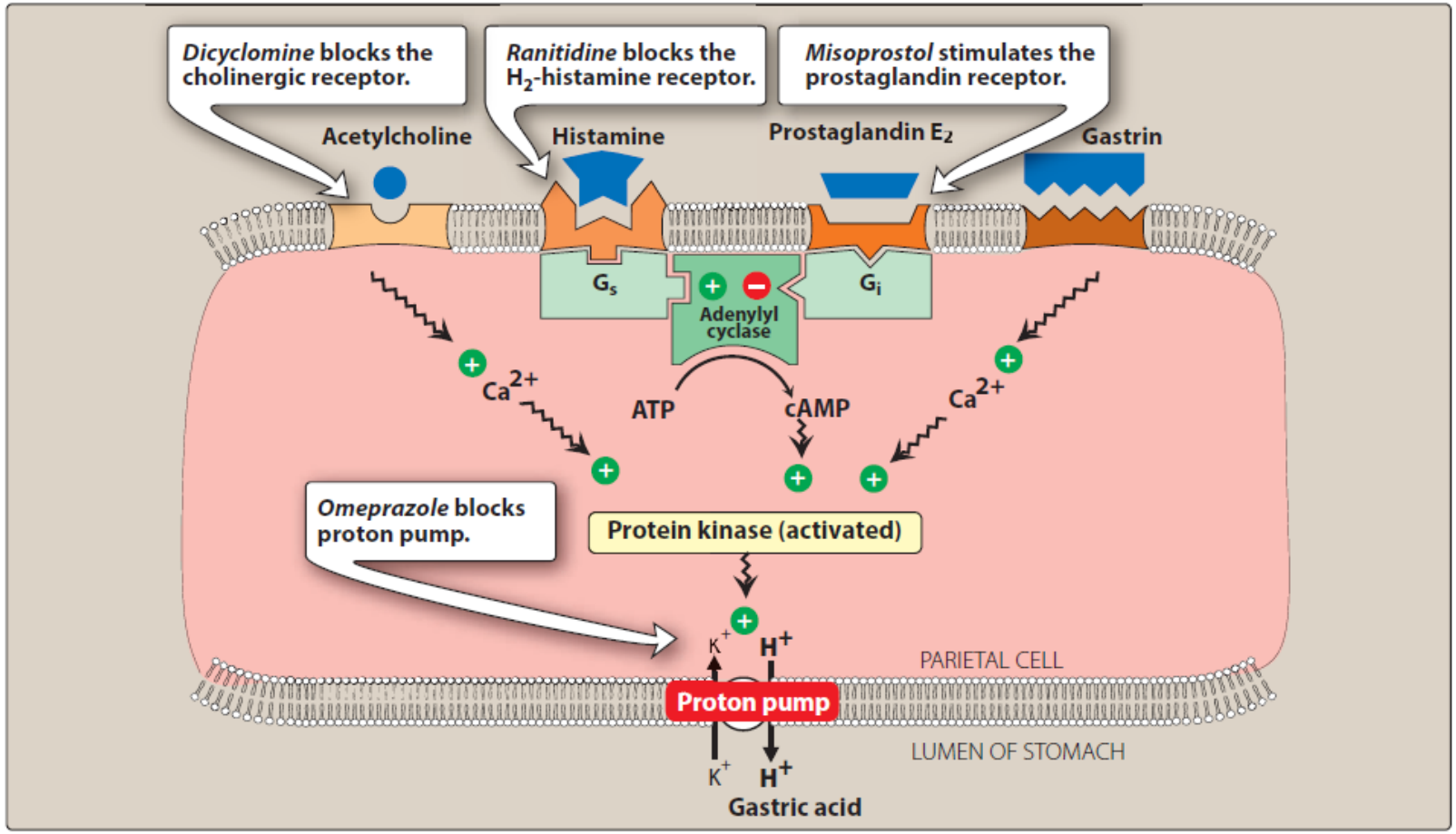
- Stopping smoking
- Avoid ulcerogenic drugs
(e.g. NSAIDs, alcohol, glucocorticosteroids)
- Reduce caffeine intake.
- Diet should be healthy
(avoid obesity, and foods that give rise to symptoms).

II- Drug treatment

Reduce intra-gastric acidity

Promote mucosal defense

Antibacterial


Dicyclomine blocks the cholinergic receptor.
Ranitidine blocks the H_2-histamine receptor.
Misoprostol stimulates the prostaglandin receptor.
Acetylcholine
Histamine
Prostaglandin E_2
Gastrin
G_s
G_i
Adenylyl cyclase
Ca^{2+}
ATP
cAMP
Ca^{2+}
Omeprazole blocks proton pump.
Protein kinase (activated)
K^+
H^+
PARIETAL CELL
Proton pump
LUMEN OF STOMACH
K^+
H^+
Gastric acid

Drug ttt of peptic ulcer

- Antacids → Acid Neutralizing Drugs
- H_2 receptor antagonist
- Proton pump inhibitor
- Muscarinic blockers

  } Gastric Antisecretory Drugs
- Enhancement of mucosal resistance → Mucosal Protective Agents
- Antibacterial Drugs

I-Antacids

- Popular antacids include **magnesium hydroxide** ($Mg[OH]_2$) and **aluminum hydroxide** ($Al[OH]_3$), calcium carbonate and sodium bicarbonate. Magnesium hydroxide has a strong laxative effect, whereas aluminum hydroxide has a constipating action. These drugs are available as **single-ingredient products** and as combined preparations.

Mechanism of action

- **Weak bases that neutralize gastric acid** and thus relieving associated pain and nausea.
- **They do not decrease the secretion of acid.**
- **Inhibit formation of pepsin** as pepsinogen converted to pepsin at acidic pH.

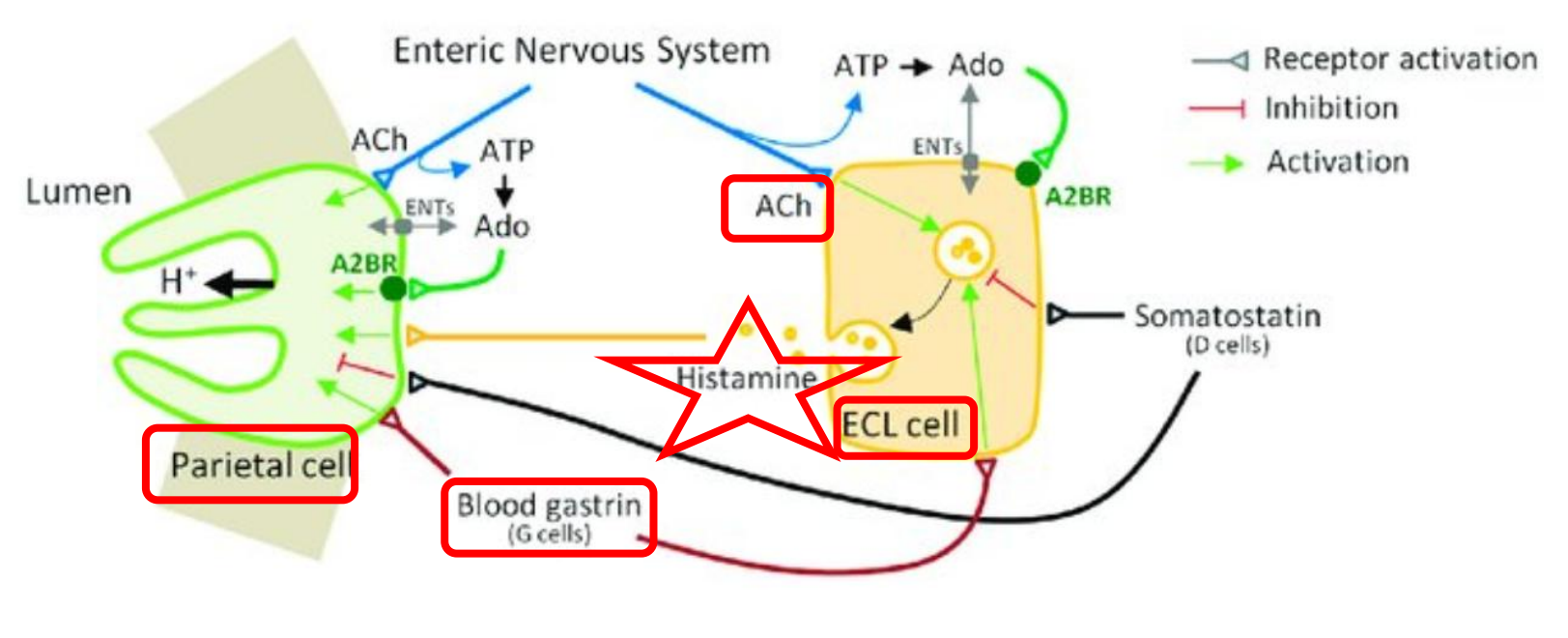
Uses

- Prevent and treat peptic ulcer.
- They are also used in the treatment of reflux esophagitis and gastritis.

ANTACID	ADVERSE EFFECT
Aluminum-containing antacids	Constipation Binds to tetracycline to form insoluble complex Antacid binds to secreted and ingested phosphate to form insolublesalts; hypophosphatemia may lead to osteomalacia and myopathy
Sodium-containing antacids	Metabolic alkalosis - Rebound hyperacidity Fluid retention - Stomach distension Hypernatremia
Magnesium-containing antacids	Hypermagnesemia, Diarrhea Hypotension
Calcium-containing antacids	Milk-alkali syndrome may lead to hypercalcemia, renal stones, and metabolic alkalosis Constipation

Aluminum and magnesium salts are most common and promote the healing of duodenal ulcers.

II-H_2-receptor antagonists



II-H_2-receptor antagonists

- **Cimetidine** and other H_2 antagonists (**ranitidine**, **famotidine**, and **nizatidine**).
- They induce **competitive inhibition at the parietal cell H2 receptor** and suppress basal and meal-stimulated acid secretion.

Mechanism of action

- Reversibly blocks H2 receptors leading to decreased H+ secretion by parietal cells.
- Very effective in inhibiting basal & nocturnal acid secretion (as it depends largely on Histamine)

Uses

- Treatment of **Gastro-esophageal reflux** (GERD),
- Treatment of **peptic ulcer disease, and non-ulcer dyspepsia.**
- Prevention of stress-related gastritis in seriously ill patients.
- Pre-anesthetic medication *(To prevent aspiration pneumonitis)*.

Side Effects [More with Cimetidine]

Extremely safe drugs

- Impairs estrogen metabolism causing **gynecomastia** in males.
- Anti-androgenic effects causing **impotence** in men.
- Hyperprolactinemia causing **galactorrhea-amenorrhea** in women.
- **Cytochrome P450 inhibition**.

III-Proton Pump inhibitors

- **Omeprazole**, **esomeprazole**, **lansoprazole**, **pantoprazole** and rabeprazole.

Mechanism of action

- Prodrugs requiring activation in acid environment.
- Irreversibly inhibits the H+/K+ ATPase pump in the stomach parietal cells & binds covalently to its extracellular domain.
- Long lasting inhibition of acid production; Acid secretion resumes only after synthesis of new molecules. Once daily dose is sufficient.
- Most effective drugs in antiulcer therapy. Profound reduction of gastric acid - elevates gastric pH significantly

N.B

They are activated in the acidic medium of the secretory parietal cell canaliculus, so:

- Given 1 hr before meals, bioavailability is reduced by food.
- Other acid suppressing (**H2 blockers-Antacids**) agents should not be co-administered.

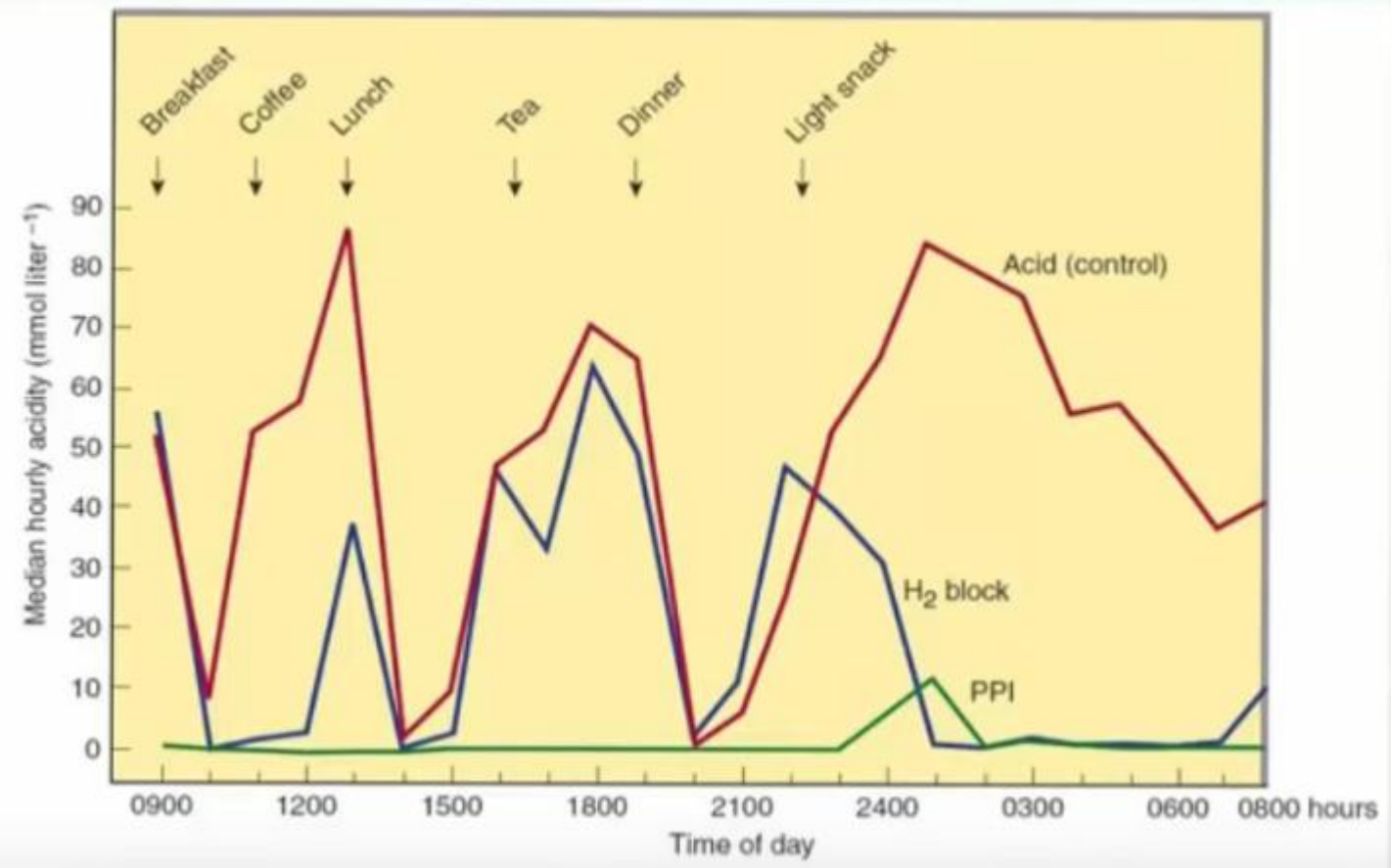
Uses

- Benign **duodenal and gastric ulcers**.
- **NSAID-associated peptic ulcer** and gastro-duodenal erosions.
- In combination with antibacterial drugs to **eradicate H. pylori**.
- **Zollinger–Ellison syndrome**.
- **Gastric acid reduction during general anesthesia**.
- Gastro-oesophageal reflux disease (**GERD**) **(**complicated or unresponsive to treatment with H2-receptor antagonists**).**

Adverse Effects

- Adverse effects of proton pump inhibitors occur infrequently and include **diarrhea, abdominal pain, and headache**.
- **Chronic treatment** with proton pump inhibitors may result in **hypergastrinemia**.
- Proton pump inhibitors may **decrease the oral bioavailability of vitamin B12** and **certain drugs that require acidity for their gastrointestinal absorption (eg, digoxin, ketoconazole)**.
- **Increased susceptibility of hip fracture** (may be related to reduced calcium absorption).
- Patients taking proton pump inhibitors may have a small increase in the risk of **respiratory** and **enteric infections**.

- Without ttt
- H2 antagonists
- Proton Pump Inhibitors



IV-Anti-cholinergics

Pirenzepine – Telenzepine

- Blocks M1 receptors on the parietal cells.
- Selectively inhibit gastric acid secretion
- No or minimal effect on gastric motility
- Less side effects of cholinergic blockade.
- Rarely used (poor efficacy- undesirable side
- effects)

Mucosal Protective Agents

Prostaglandin analogues
(Misoprostol)

- An analog of PGE1, misoprostol **increases mucosal protection through increasing mucous and bicarbonate production and inhibits gastrin release, pepsin and acid secretion.**

Uses

- Prevention of NSAID gastric ulcers.

Adverse Effects

- Diarrhea
- **Abdominal cramps (sever colicky pain).**
- Abortion
- Exacerbate IBD

Colloidal bismuth

Bismuth subcitrate/ subsalicylate

Mechanism of action:

- They promote **mucus and bicarbonate production.**
- They bind to proteins in the necrotic ulcer crater forming **a barrier** to the diffusion of acid (physical mechanism).
- Decrease activity of pepsin
- They have a significant **antibacterial effect on H. pylori.**

Uses:

- Triple therapy for eradication of H. pylori.
- Peptic ulcer.
- Traveller's diarrhea

Adverse Effects:

- Black stool.
- Constipation
- Teeth discoloration.
- Bismuth toxcity: Encephalopathy (in renal dysfunction).

Sucralfate

Complex of **Sulfated sucrose + aluminium hydroxide**

Mechanism of action

1. **In acidic pH, sucralfate dissociates into its components.**
2. **The negatively charged sulfated sucrose binds with positively charged protein molecules found in damaged mucosa** and adheres strongly to those proteins found at the base of the ulcer. Thus, it **forms a** viscous gel that adheres to ulcer crater **(Coat over the ulcer).**
3. Promote ulcer healing.
4. Inhibition of pepsin
5. Stimulation of mucosal protective mechanisms (PG& mucous and bicarbonates secretion).

Adverse Effects:

- Constipation
- Dry mouth.
- Interferes with absorption of some drugs tetracycline, theophyline, Tricyclic antidepressant.

Eradication of H. Pylori

- **Antibiotics commonly used are:**
- **Clarithromycin,**
- **Tetracycline,**
- **Amoxicillin**

Metronidazole: used often due to bacterial resistance to amoxicillin and tetracycline, or due to penicillin allergy

- **Triple therapy** for 14 days is considered the ttt of choice…….PPI + clarithromycin and amoxicillin
- **Quadriple therapy**:

Bismuth subsalicylate, Metronidazole, tetracycline, PPI

- **After completion** of H. pylori eradication therapy, PPI is continued for 4-6 weeks

Aluminum Hydroxide + Magnesium Hydroxide

Cimetidine (Tagamet®)

Ranitidine (Zantac®)

Omeprazole (Pepzol®)

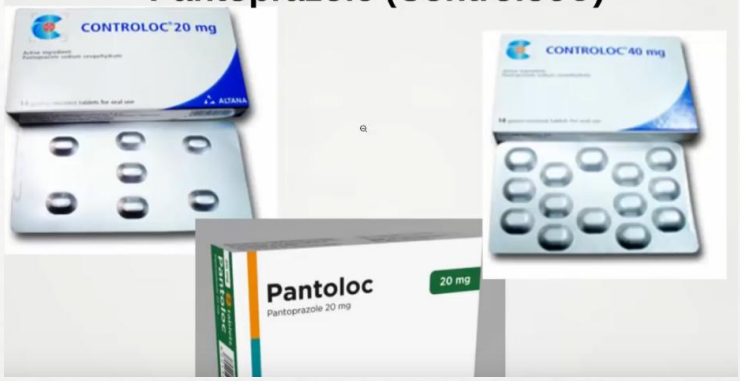
Esomeprazole (Nexium®)

Lansoprazole (Lanzor®)

Dexlansoprazole (Dexilant®)

Granule 1 begins releasing medicine within an hour of taking DEXILANT.
Granule 2 provides a second release of medicine around 4-5 hours later.

Pantoprazole (Controloc®)



Rabeprazole (Pariet®)

THANK
YOU!